

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 2.2: Growth and Development in Plants — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 2.2: Growth and Development in Plants
Driving Question	DRIVING QUESTION: What causes the differences in germination, emergence and growth of maize and bean plants?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Farmer's Puzzle: Why Do Maize and Beans Behave Differently After the Same Rain?	Students looked at a case of a farmer in Kakamega who planted maize and beans on the same day, and the same rain fell on both. Maize emerged in about 5-7 days with the seed staying underground, beans emerged later and carried the seed leaves above the soil, and a few seeds of both did not emerge at all. Teacher note: record the exact differences students name on the board (speed, appearance, number emerging) and do not correct them yet.	The same rain and same planting day do not give the same result for every seed. Students learned that they do not yet know what causes these differences. They generated questions for the Driving Question Board (DQB) about seed structure, water, dormancy, temperature, and the roles of hormones. Teacher note: accept all reasonable questions and sort them into themes (seed, conditions, internal control, measurement).	Teacher reference: at this stage students give only initial ideas, e.g. 'beans are bigger', 'some seeds are bad' or 'maize likes rain more'. Treat these as preconceptions to be revisited. The anchoring phenomenon is that identical conditions give different outcomes, which implies that factors inside the seed and its needs vary. Use WAIT TIME of at least 5 seconds after asking 'What might be different between a maize seed and a bean seed?' and cold-call 3 students. Build the first initial model as a sketch of seed + soil + rain with question marks for unknowns.
Lesson 2 Meaning of Growth and Development; Structure of Monocot and Dicot Seeds	Students soaked maize grains and bean seeds, dissected them and drew what they saw using hand lenses. Maize shows one cotyledon, a large endosperm, a coleoptile and coleorhiza, and the pericarp fused to the testa. Beans show two large cotyledons, a testa, micropyle, hilum, plumule and radicle, and no endosperm. Teacher note: bean seeds split easily when soaked overnight, while maize needs a sharp blade cut along the length.	Growth is a permanent increase in size, dry mass or cell number, while development is the change in form and function as cells differentiate. Maize is a monocot with one cotyledon and an endosperm store, and beans are dicots with two cotyledons that hold the food reserve. Students also learned to label the parts of both seeds correctly.	Teacher reference: the structural differences in where food is stored (endosperm in maize, cotyledons in beans) and in how the embryo is protected (coleoptile and coleorhiza in maize) are the first explanation for the different behaviour of ugali maize and githeri beans after planting. A common misconception to address is that growth and development are the same thing. Use a simple contrast: a seedling getting taller is growth, a root cell becoming a root hair cell is development.

			Add seed structures to the DQB answers and the class model.
Lesson 3 Seed Dormancy: Why Do Some Healthy-Looking Seeds Refuse to Germinate?	Students compared untreated bean or acacia seeds with seeds that were scarified (nicked with a razor or rubbed with sandpaper), soaked in warm water, or left untreated. Treated seeds took up water and swelled while many untreated seeds with hard coats stayed unchanged. Teacher note: use acacia or Calliandra seeds from the school compound if available, since they show dormancy very clearly.	Seed dormancy is a resting state in which a living seed fails to germinate even when conditions seem suitable. Causes include a hard impermeable seed coat, immature embryos, chemical inhibitors such as abscisic acid, and a need for light or cold. It is broken by scarification, soaking, stratification, leaching, light exposure or hormone changes. Dormancy helps seeds survive unfavourable seasons.	Teacher reference: this explains why a few seeds in the farmer's field never emerged even though the rain was the same. They were alive but dormant, not dead. Stress the difference between dormant and non-viable seeds, and note that a farmer in the Rift Valley or Kericho may nick or soak tree seeds before sowing a nursery. Ask 'How can you tell a dead seed from a dormant seed?' and give 5 seconds of WAIT TIME. Update the DQB with dormancy and add it to the model as a 'gate' before germination.
Lesson 4 Germination: Process, Types and Conditions Necessary for Germination in Maize and Bean Seeds	Students set up controlled experiments with maize and beans in test tubes or on cotton wool with a control, no water, no oxygen (boiled water with oil layer), and cold or warm conditions. Only the seeds with water, oxygen and a suitable temperature germinated, and maize and beans showed different seedling patterns. Teacher note: ensure the control is labelled and that only one variable changes in each set-up.	Seeds need water, oxygen and a suitable temperature to germinate, and they must be alive (viable) and not dormant. Germination starts with water uptake (imbibition), enzymes are activated, stored food is digested, and the radicle emerges first. Beans show epigeal germination, where cotyledons are lifted above the ground, while maize shows hypogeal germination, where the food store stays below the ground.	Teacher reference: this explains the farmer's puzzle directly, because bean seed leaves appear above the soil (epigeal) while maize shoots appear while the grain remains underground (hypogeal). Water, oxygen and temperature were the same for both, but the structures that respond differ. Link to waterlogged soils around Lake Victoria, which cut oxygen and hinder germination. Revisit the model: add imbibition, enzymes, radicle first, and the two types of germination.
Lesson 5 Primary and Secondary Growth: How Do Plants Show Growth?	Students examined prepared slides or diagrams of root and shoot tips and cross-sections of young and old stems, and compared a maize seedling stem with an older bean or tree stem such as grevillea or cypress. They noticed that tips show dividing cells, while older dicot stems have a ring of cambium and woody tissue. Teacher note: if there are no slides, use clear photographs.	Primary growth is an increase in length produced by apical meristems at shoot and root tips, while secondary growth is an increase in thickness produced by lateral meristems (vascular cambium and cork cambium). Monocots such as maize generally show only primary growth, while many dicots, especially woody ones, show secondary growth.	Teacher reference: this helps explain why a maize stalk grows tall but does not form annual rings, while a tree dicot thickens over the years. Beans are herbaceous dicots and show limited secondary growth. A common misconception is that all plants thicken with rings, so ask students to justify with evidence from the sections. Add meristems to the model as 'growth engines' and update the DQB.
Lesson 6 Where Does a Maize or Bean Plant Grow? Measuring Growth and Regions of Growth in Seedlings	Students marked equal intervals (for example every 2 mm) with ink on young maize or bean roots and measured them again after 24-48 hours. The marks nearest the tip spread the furthest apart, while those close to the seed barely moved. Teacher note: use fine thread and waterproof ink, and keep roots moist to avoid damage.	Growth in length is concentrated in specific regions, namely the apical meristems at the root and shoot tips followed by the zone of elongation, while stems thicken through lateral meristems. The regions are cell division, cell elongation and cell differentiation (maturation).	Teacher reference: this shows that growth is not uniform along the plant, which explains why maize seedlings and bean seedlings can differ in height and form even when conditions are the same. Use the data to answer 'Where exactly is the new growth happening?' and cold-call 3 students to share measurements. Add regions of

			growth to the class model and tie to the DQB question on how plants grow.
Lesson 7 Plant Hormones: What Controls Growth in Maize and Bean Seedlings?	Students watched or modelled phototropism with maize coleoptiles or bean seedlings placed near a window or covered with a box with a slit, and saw seedlings bending toward light. They also studied data on seeds treated with different hormone solutions or on the ripening of bananas near ripe fruit. Teacher note: allow 3-5 days for bending to appear.	Hormones are chemical messengers that coordinate germination, dormancy, shoot and root growth, branching and fruit ripening. They work together with environmental factors such as light, gravity and water. Key hormones are auxins, gibberellins, cytokinins, abscisic acid and ethylene.	Teacher reference: hormones are the internal control that was missing in earlier lessons. They explain how a seed 'knows' when to germinate, why stems bend to light and why mangoes and bananas ripen together in a sack in a market in Mombasa. Emphasise that hormones act in low concentrations and interact, rather than one hormone acting alone. Add 'chemical messengers' to the model, linked to dormancy and growth.
Lesson 8 Measuring Growth: Seedling Height, Stem Width and Growth Curves of Maize and Beans	Students measured the height and stem width of maize and bean seedlings over several days, recorded data in a table, and plotted graphs. The curves were S-shaped, with slow early growth, a fast middle phase and a levelling off. Teacher note: encourage the use of a ruler and thread and mark the same point on the stem each time.	Growth can be measured as height, stem width, mass or leaf area. A growth curve shows that growth rate changes over time (lag, exponential and stationary phases). Fresh mass changes with water content, whereas dry mass is a more reliable measure of true growth.	Teacher reference: the different shapes and timings of maize and bean curves give quantitative evidence for the differences in the driving question. Students should use data, e.g. 'maize grew X cm per day between day 5 and day 10', as evidence. Check that axes, units and labels are correct. Add the growth curve to the model and review with 'Why does growth slow down later?'
Lesson 9 Role of Hormones in Plant Growth and Development: Why Do Maize and Bean Plants Grow Differently?	Students analysed case data and experiments on hormones, such as decapitated shoots, auxin applied to cuttings, gibberellin on dwarf plants and ethylene on fruits. They saw that removing a shoot tip increased side branching and that treated cuttings rooted more readily. Teacher note: relate to farmers pruning tea in Kericho or pinching out coffee and beans to increase branching.	Hormones are chemical messengers that control germination, dormancy, cell elongation, branching, rooting and fruit ripening. Auxin from the shoot tip causes apical dominance and cell elongation, gibberellins promote germination and stem elongation, cytokinins promote cell division and branching, abscisic acid maintains dormancy and closes stomata, and ethylene ripens fruit.	Teacher reference: differences in the balance and timing of hormones, together with differences in structure, help explain why maize and beans grow differently, for example in height, branching and leaf arrangement. Students should explain with cause-and-effect chains, not just list hormones. Use cold-calls (3 students) and WAIT TIME of 5 seconds. Add hormone roles to the model and mark off DQB questions that are now answered.
Lesson 10 Synthesis and Application: Explaining the Farmer's Maize and Beans and Planning the Tree Nursery	Students revisited the farmer's puzzle from Lesson 1 and used all their evidence (seed structure, dormancy tests, germination experiments, growth measurements and hormone data) to explain the results. They then applied it to planning a tree nursery, deciding how to treat seeds, water, shade and space seedlings. Teacher note: ask groups to present using a final model and evidence list.	Germination, emergence and growth depend on seed structure and food stores, dormancy and its breaking, water, oxygen and suitable temperature, meristems and regions of growth, and the balance of hormones. These factors act together, and maize and beans differ in several of them at the same time.	Teacher reference: a full answer links structure (monocot vs dicot, endosperm vs cotyledons, hypogeal vs epigeal), dormancy (some seeds not ready), conditions (water, oxygen, temperature, soil), growth patterns (meristems, curves) and hormones (internal control). In the nursery, students may suggest scarifying hard-coated seeds, using well-drained soil, shading seedlings, and pruning for branching. Check the final

			models for evidence-based reasoning rather than memorised lists, and close by returning to the DQB to show which questions have been answered and which remain for further inquiry.
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